

Week 3: Final Report

MLFQ Scheduler for xv6-RISC-V

Priority Boosting, Starvation Prevention, and Complete Validation

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1 Project Overview

1.1 Introduction

This final report documents the complete implementation of a Multi-Level Feedback Queue (MLFQ) scheduler for xv6-RISC-V. The project replaces xv6's default round-robin scheduler with a sophisticated priority-based scheduler that:

- Automatically adapts to process behavior
- Favors interactive and I/O-bound processes
- Prevents starvation through periodic priority boosting
- Maintains fairness across all process types

1.2 Project Timeline

Week	Focus	Deliverables
1	Design & Foundation	Data structures, syscalls, design document
2	Core Implementation	Scheduler logic, demotion, test programs
3	Boosting & Validation	Priority boost, starvation prevention, final tests

Table 1: Project Timeline

2 Final MLFQ Configuration

2.1 Queue Structure

Queue	Priority	Time Quantum	Process Type
Q0	Highest	2 ticks (200ms)	Interactive, I/O-bound
Q1	High	4 ticks (400ms)	Mixed workloads
Q2	Medium	8 ticks (800ms)	Compute-light
Q3	Lowest	16 ticks (1.6s)	CPU-bound, batch

Table 2: Final Queue Configuration

2.2 Scheduling Parameters

```

1 #define NMLFQ 4           // Number of priority queues
2 #define BOOST_INTERVAL 30 // Boost all processes every 30
   ticks
3
4 int mlfq_time_quanta[NMLFQ] = {2, 4, 8, 16}; // Ticks per queue
5 uint64 last_boost_time = 0; // Track time since last boost

```

Listing 1: MLFQ Parameters (kernel/proc.c)

3 Priority Boosting Implementation

3.1 Starvation Problem

Without priority boosting, a CPU-bound process demoted to Q3 could starve if:

- Higher-priority processes continuously arrive
- The scheduler never reaches Q3
- The process waits indefinitely

3.2 Solution: Periodic Boosting

Every 30 ticks (3 seconds), all processes are boosted back to Q0:

```
1 void
2 scheduler(void)
3 {
4     struct proc *p;
5     struct cpu *c = mycpu();
6
7     c->proc = 0;
8     for(;;){
9         intr_on();
10        intr_off();
11
12        // Check if it's time for priority boosting
13        acquire(&tickslock);
14        uint64 current_ticks = ticks;
15        if(current_ticks - last_boost_time >= BOOST_INTERVAL) {
16            last_boost_time = current_ticks;
17            release(&tickslock);
18            boost_all_priorities(); // BOOST ALL PROCESSES TO Q0
19        } else {
20            release(&tickslock);
21        }
22
23        // ... MLFQ scheduling logic ...
24    }
25 }
```

Listing 2: Priority Boosting in Scheduler (kernel/proc.c)

3.3 Boost Function

```
1 void
2 boost_all_priorities(void)
3 {
4     struct proc *p;
5     int boosted_count = 0;
```

```

6
7  for(p = proc; p < &proc[NPROC]; p++) {
8      acquire(&p->lock);
9      if(p->state == RUNNABLE || p->state == RUNNING || p->state ==
10         SLEEPING) {
11          if(p->priority != 0 || p->time_slices != 0) {
12              boosted_count++;
13          }
14          p->priority = 0;           // Reset to highest priority
15          p->time_slices = 0;       // Reset time slice counter
16      }
17      release(&p->lock);
18  }

```

Listing 3: boost_all_priorities Function (kernel/proc.c)

3.4 boostproc System Call

Added a manual boost syscall for testing:

```

1  uint64
2  sys_boostproc(void)
3  {
4      extern void boost_all_priorities(void);
5
6      boost_all_priorities();
7
8      return 0;
9  }

```

Listing 4: boostproc System Call (kernel/sysproc.c)

4 Extended Boost Test

4.1 boostlong.c Test Program

A comprehensive test that demonstrates the complete MLFQ cycle:

```

1  #include "kernel/types.h"
2  #include "kernel/stat.h"
3  #include "user/user.h"
4
5  void pure_cpu_work(int iterations) {
6      volatile int dummy = 0;
7      for(int i = 0; i < iterations; i++) {
8          for(long j = 0; j < 500000; j++) {
9              dummy = dummy + j;
10             dummy = dummy % 1000000;
11         }
12     }

```

```
13 }
14
15 int main(int argc, char *argv[])
16 {
17     struct procinfo info;
18     int last_priority = -1;
19     int boost_count = 0;
20     int phase = 0;
21
22     printf("
23     =====\n");
24     printf("          MLFQ BOOST TEST - EXTENDED VERSION\n");
25     printf("
26     =====\n");
27     printf("PHASE 1: STARTUP OVERHEAD ANALYSIS\n");
28     printf("
29     =====\n");
30     int t0 = uptime();
31     getprocinfo(&info);
32     int t1 = uptime();
33     printf("After getprocinfo: uptime %d->%d, slices=%d, queue=Q%d\n",
34           t0, t1, info.time_slices, info.priority);
35
36     printf("\nPHASE 2: WAIT FOR BOOST TO RESET\n");
37     printf("
38     =====\n");
39     int waited = 0;
40     while(1) {
41         pure_cpu_work(5);
42         waited++;
43         getprocinfo(&info);
44         if(info.priority == 0 && info.time_slices == 0 && waited >
45            10) {
46             printf("Got boosted! Starting clean measurement now.\n\n");
47             break;
48         }
49         if(waited > 200) {
50             printf("Timeout waiting for boost, starting anyway.\n\n");
51             break;
52         }
53     }
54     printf("
55     =====\n");
```

```

=====
n");
54 printf("PHASE 3: CLEAN MLFQ DEMONSTRATION\n");
55 printf("
=====
n");
56 printf("Time Quanta: Q0=2, Q1=4, Q2=8, Q3=16 slices\n");
57 printf("Boost Interval: Every 30 ticks all processes -> Q0\n\n"
);
58
59 printf("
+-----+-----+-----+-----+\n")
;
60 printf("| SLICE | QUEUE | CHANGE | DESCRIPTION
|\n");
61 printf("
+-----+-----+-----+-----+\n")
;
62
63 getprocinfo(&info);
64 last_priority = info.priority;
65 printf("| %d | Q%d | START | Fresh start highest
priority |\n",
66 info.time_slices, info.priority);
67
68 while(phase < 400) {
69 pure_cpu_work(1);
70 phase++;
71
72 getprocinfo(&info);
73
74 if(info.priority != last_priority) {
75 if(info.priority > last_priority) {
76 // Demotion
77 if(last_priority == 0) {
78 printf("| %d | Q%d | DEMOTE | Used 2 slices ->
Q1 |\n",
79 info.time_slices, info.priority);
80 } else if(last_priority == 1) {
81 printf("| %d | Q%d | DEMOTE | Used 4 slices ->
Q2 |\n",
82 info.time_slices, info.priority);
83 } else if(last_priority == 2) {
84 printf("| %d | Q%d | DEMOTE | Used 8 slices ->
Q3 |\n",
85 info.time_slices, info.priority);
86 }
87 } else {
88 // Boost detected!
89 boost_count++;
90 printf("
=====

```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
;
91     printf("|    %d    | Q%d    | BOOST    | *** AUTOMATIC BOOST
%d ***    |\n",
92             info.time_slices, info.priority, boost_count);
93     printf("
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
;
94     }
95     }
96
97     last_priority = info.priority;
98
99     if(boost_count >= 3) break;
100 }
101
102 printf("
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
;
103
104 printf("\n
=====
n");
105 printf("
TEST SUMMARY\n");
106 printf("
=====
n");
107 printf("    Boost Events: %d automatic boosts observed\n",
boost_count);
108 printf("
-----
n");
109 if(boost_count >= 2) {
110     printf("    SUCCESS! MLFQ working correctly:\n");
111     printf("    - Q0: 2 slices -> demote to Q1\n");
112     printf("    - Q1: 4 slices -> demote to Q2\n");
113     printf("    - Q2: 8 slices -> demote to Q3\n");
114     printf("    - Q3: stays until boost (every 30 ticks)\n");
115     printf("    - BOOST: all processes return to Q0\n");
116 }
117 printf("
=====
n\n");
118
119 exit(0);
120 }

```

Listing 5: Extended Boost Test (user/boostlong.c)

4.2 boostlong Test Output


```

1 $ boostlong
2 =====
3     MLFQ BOOST TEST - EXTENDED VERSION
4 =====
5
6 PHASE 1: STARTUP OVERHEAD ANALYSIS
7 -----
8 After getprocinfo: uptime 163->163, slices=0, queue=Q0
9
10 PHASE 2: WAIT FOR BOOST TO RESET
11 -----
12 Got boosted! Starting clean measurement now.
13
14 =====
15 PHASE 3: CLEAN MLFQ DEMONSTRATION
16 =====
17 Time Quanta: Q0=2, Q1=4, Q2=8, Q3=16 slices
18 Boost Interval: Every 30 ticks all processes -> Q0
19
20 +-----+-----+-----+-----+
21 | SLICE | QUEUE | CHANGE | DESCRIPTION |
22 +-----+-----+-----+-----+
23 | 0 | Q0 | START | Fresh start highest priority |
24 | 0 | Q1 | DEMOTE | Used 2 slices -> Q1 |
25 | 0 | Q2 | DEMOTE | Used 4 slices -> Q2 |
26 | 0 | Q3 | DEMOTE | Used 8 slices -> Q3 |
27 +-----+-----+-----+-----+
28 | 0 | Q0 | BOOST | *** AUTOMATIC BOOST #1 *** |
29 +-----+-----+-----+-----+
30 | 0 | Q1 | DEMOTE | Used 2 slices -> Q1 |
31 | 0 | Q2 | DEMOTE | Used 4 slices -> Q2 |
32 | 0 | Q3 | DEMOTE | Used 8 slices -> Q3 |
33 +-----+-----+-----+-----+
34 | 0 | Q0 | BOOST | *** AUTOMATIC BOOST #2 *** |
35 +-----+-----+-----+-----+
36 | 0 | Q1 | DEMOTE | Used 2 slices -> Q1 |
37 | 0 | Q2 | DEMOTE | Used 4 slices -> Q2 |
38 | 0 | Q3 | DEMOTE | Used 8 slices -> Q3 |
39 +-----+-----+-----+-----+
40 | 0 | Q0 | BOOST | *** AUTOMATIC BOOST #3 *** |
41 +-----+-----+-----+-----+
42 +-----+-----+-----+-----+
43
44 =====
45     TEST SUMMARY
46 =====
47 Boost Events: 3 automatic boosts observed
48 -----
49 SUCCESS! MLFQ working correctly:
50 - Q0: 2 slices -> demote to Q1

```

```

51 - Q1: 4 slices -> demote to Q2
52 - Q2: 8 slices -> demote to Q3
53 - Q3: stays until boost (every 30 ticks)
54 - BOOST: all processes return to Q0
55 =====

```

Listing 6: boostlong Terminal Output

5 Complete System Architecture

5.1 Data Flow

1. **Process Creation:** New process starts at Q0 with `time_slices=0`
2. **Timer Interrupt:** On each tick, `time_slices++`
3. **Demotion Check:** If `time_slices >= quantum[priority]`, demote
4. **Scheduler:** Always picks from highest non-empty queue
5. **Boost Check:** Every 30 ticks, all processes $\rightarrow$ Q0

5.2 Files Modified

File	Modification
kernel/proc.h	Added MLFQ fields to <code>struct proc</code>
kernel/proc.c	MLFQ scheduler, boost function, initialization
kernel/trap.c	Time slice tracking, demotion logic
kernel/sysproc.c	<code>getprocinfo</code> , <code>boostproc</code> syscalls
kernel/syscall.h	Syscall numbers
kernel/syscall.c	Syscall registration
user/user.h	User-space declarations
user/usys.pl	Syscall stubs
user/*.c	Test programs
Makefile	Build configuration

Table 3: Complete File List

6 Complete Test Results

6.1 Test Summary

6.2 Key Observations

1. **Demotion Works:** CPU-bound processes correctly demote through all queues
2. **I/O Priority:** Processes that yield frequently maintain high priority

Test	Expected Behavior	Result
procinfo	Returns correct process info	PASS
cpubound	Demotes Q0 $\rightarrow$ Q3	PASS
iobound	Stays in Q0	PASS
mlfqtest	I/O gets priority over CPU	PASS
boostlong	Shows 3+ boost cycles	PASS

Table 4: Final Test Results

3. **Boosting Works:** All processes are boosted every 30 ticks
4. **Starvation Prevented:** No process is starved indefinitely
5. **Fairness:** Mixed workloads demonstrate correct priority differentiation

7 Performance Characteristics

7.1 Responsiveness

- Interactive processes stay in Q0/Q1 with 2-4 tick quanta
- Response time: 200-400ms for I/O-bound processes
- System feels responsive even under CPU load

7.2 Throughput

- CPU-bound processes get longer quanta at Q3 (16 ticks)
- Reduced context switching for compute-intensive work
- Good overall CPU utilization

7.3 Fairness

- Boosting every 30 ticks ensures no starvation
- All processes get periodic high-priority access
- Round-robin within each queue ensures fairness among peers

8 Conclusion

8.1 Achievements

Successfully implemented 4-level MLFQ scheduler

Automatic demotion based on CPU usage

Priority boosting prevents starvation

I/O-bound processes receive preferential treatment

Comprehensive test suite validates all features

System remains stable and responsive

8.2 MLFQ Rules Implemented

1. **Rule 1:** If $\text{Priority}(A) > \text{Priority}(B)$, A runs
2. **Rule 2:** If $\text{Priority}(A) = \text{Priority}(B)$, round-robin
3. **Rule 3:** New processes start at highest priority (Q0)
4. **Rule 4:** If process uses full quantum, demote
5. **Rule 5:** After boost interval, all processes $\rightarrow$ Q0

8.3 Future Improvements

Potential enhancements for future work:

- Gaming prevention (detect processes that reset timers)
- Configurable queue parameters
- Per-process priority adjustment syscall
- Multi-core MLFQ scheduling
- Real-time priority class support

9 Appendix: System Call Reference

9.1 getprocinfo

```
1 int getprocinfo(struct procinfo *info);
```

Description: Retrieves scheduling information for the calling process.

Returns: 0 on success, -1 on failure.

Structure:

```
1 struct procinfo {  
2     int pid;           // Process ID  
3     int state;         // Process state  
4     int priority;      // Current queue (0-3)  
5     int time_slices;   // Slices used in current queue  
6 };
```

9.2 boostproc

```
1 int boostproc(void);
```

Description: Manually triggers priority boosting for all processes.

Returns: 0 on success.

Use Case: Testing and debugging priority boost behavior.